

ORIGINAL RESEARCH

Hospital Readmission After Total Hip & Knee Arthroplasty: The Role of Diabetes Management Strategy

Study 1 — Version 2 | April 2026 | Target: *Journal of Arthroplasty*

Method: Firth penalised logistic regression, bootstrap 95% CI (1,000 iter). Reference arm: Metformin. GLP-1 agonists merged into OtherDM arm.

1. Study Overview

This retrospective cohort study investigates whether the class of pre-surgical diabetes pharmacotherapy is associated with 30- and 90-day hospital readmission following elective total hip arthroplasty (THA) or total knee arthroplasty (TKA). Patients with type 2 diabetes mellitus (T2DM) were drawn from a large electronic health record (EHR) database and assigned to one of four treatment arms based on their drug exposure in the 365 days before surgery: **Metformin** (reference), **Insulin**, **OtherDM** (GLP-1 receptor agonists, SGLT-2 inhibitors, DPP-4 inhibitors, sulfonylureas, and other agents), and **DietControlled** (no pharmacological treatment).

Firth penalised logistic regression was selected to address the separation problem common in analyses with small treatment arms and binary outcomes. All models were adjusted for age, BMI, HbA1c, diabetes duration, surgery type, hypertension, hyperlipidaemia, coronary artery disease, depression, and baseline inpatient visit count. Results are cross-checked with generalised linear models (GLM) using HC3 robust standard errors.

Key Design Decisions

Decision	Rationale
Reference arm: Metformin	First-line T2DM therapy with the largest pharmacological arm (n=125). Provides a clinically meaningful comparator.
GLP-1 merged into OtherDM	GLP-1 users were distributed across arms (n=218 total). Merging preserves power while grouping non-insulin, non-metformin agents.
Firth penalised logistic regression	Corrects quasi-complete separation bias expected with the small OtherDM arm (n=47) and binary outcomes.
Bootstrap 95% CI (1,000 iter.)	More reliable than asymptotic CIs with unequal arm sizes and low event rates.
365-day exposure window	Captures the predominant peri-operative treatment strategy. Confirmed stable in the 180-day sensitivity analysis (8A).
Co-primary outcomes: 30d and 90d readmission	Both windows are standard in arthroplasty literature; 30d captures immediate post-surgical complications, 90d captures the full recovery period.

2. Cohort Construction

Starting from 1,953 persons, 49 patients with T1DM and 128 with insufficient baseline observation were excluded. The final analytic cohort comprised **976 patients** across four arms. Surgery type was balanced: ~60% TKA and ~40% THA in each

arm.

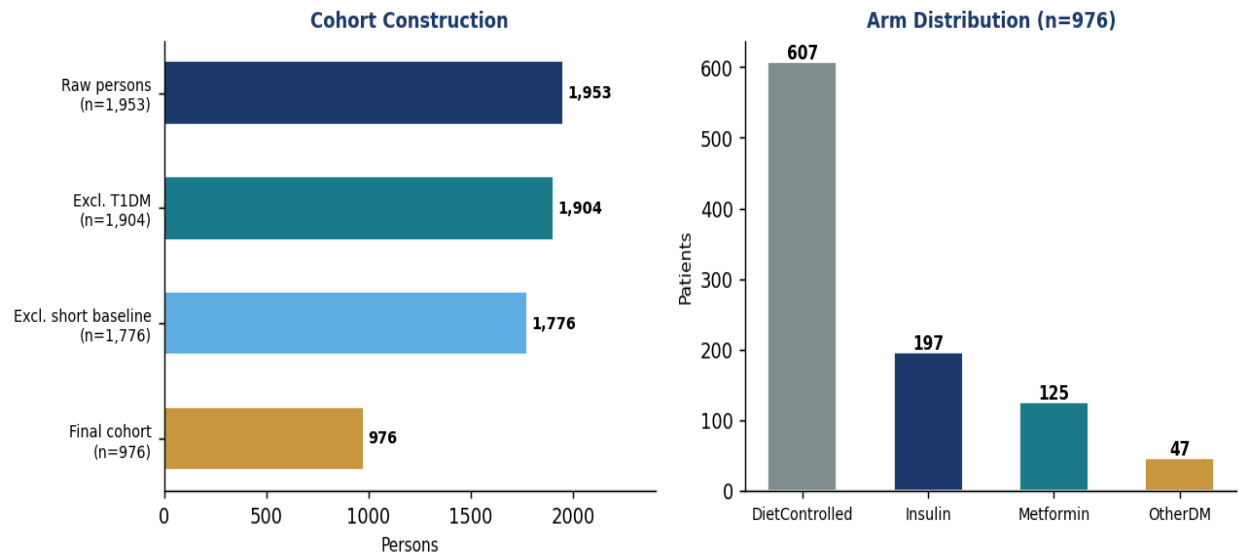


Figure 1. Left: Cohort exclusion funnel. Right: Final arm distribution.

3. Study Variables — Complete Reference

The table below describes every variable considered in this analysis, including its type, role, availability, and precise operational definition.

Variable	Type	Role	Availability	Definition / Notes
90-Day Readmission (co-primary)	Binary	Outcome	100%	Any inpatient hospital admission occurring within 90 days of the index THA/TKA, regardless of cause (all-cause). Identified from visit records with inpatient encounter type.
30-Day Readmission (co-primary)	Binary	Outcome	100%	Any inpatient admission within 30 days of surgery (all-cause). Captures immediate post-operative complications and short-term surgical failure.
90-Day Readmission or ED (secondary)	Binary	Outcome	100%	Composite: any inpatient readmission OR any emergency department visit within 90 days. Captures the full spectrum of unplanned acute-care utilisation including ED visits not resulting in admission.
DM Treatment Arm	Categorical (4 levels)	Main Predictor	100%	Assigned from drug dispensation claims in the 365 days prior to surgery. Hierarchy: Insulin > OtherDM > Metformin > DietControlled. GLP-1 agonists merged into OtherDM.
Age at surgery	Continuous (yrs)	Covariate	100%	Patient age on the date of the index procedure. Included as a linear term.
BMI	Continuous (kg/m2)	Covariate	54.5% (n=532)	Most recent pre-surgery body mass index from vital-sign measurement records.
HbA1c	Continuous (%)	Covariate	46.9% (n=458)	Most recent pre-surgery glycated haemoglobin. Missing values handled by available-case analysis per model.
Diabetes Duration	Continuous (yrs)	Covariate	100%	Time from the earliest diabetes diagnosis code to the surgery date. Proxy for disease burden.
Surgery Type	Binary (THA/TKA)	Covariate	100%	Index procedure type. TKA in 61-63% of each arm; included to adjust for differential complication profiles.
Hypertension	Binary	Covariate	100%	Presence of at least one ICD code for hypertension in the 365-day baseline window.
Hyperlipidaemia	Binary	Covariate	100%	Presence of at least one ICD code for hyperlipidaemia in the baseline window.
CAD	Binary	Covariate	100%	Coronary artery disease flag from condition records in the baseline period.
Depression	Binary	Covariate	100%	Depression diagnosis present in the baseline window. Included as a known correlate of healthcare utilisation.
Baseline Inpatient Visits	Continuous (count)	Covariate	100%	Number of inpatient encounters in the 365-day baseline period. Proxy for overall comorbidity burden.
Outpatient Visits	Continuous (count)	Descriptive	100%	Outpatient encounter count in baseline. Reported in Table 1; not included as model covariate.
ED Visits	Continuous (count)	Descriptive	100%	Emergency department visit count in baseline. Reported in Table 1 only.

Variable	Type	Role	Availability	Definition / Notes
GLP-1 Agonist Use	Binary	Note	100%	Identified within each arm (n=218 total: 86 DietControlled, 82 Insulin, 38 Metformin, 12 OtherDM). Merged into OtherDM arm in v2 analysis.

Blue rows = outcomes. Orange row = main predictor. White/light-grey rows = covariates and descriptive variables.

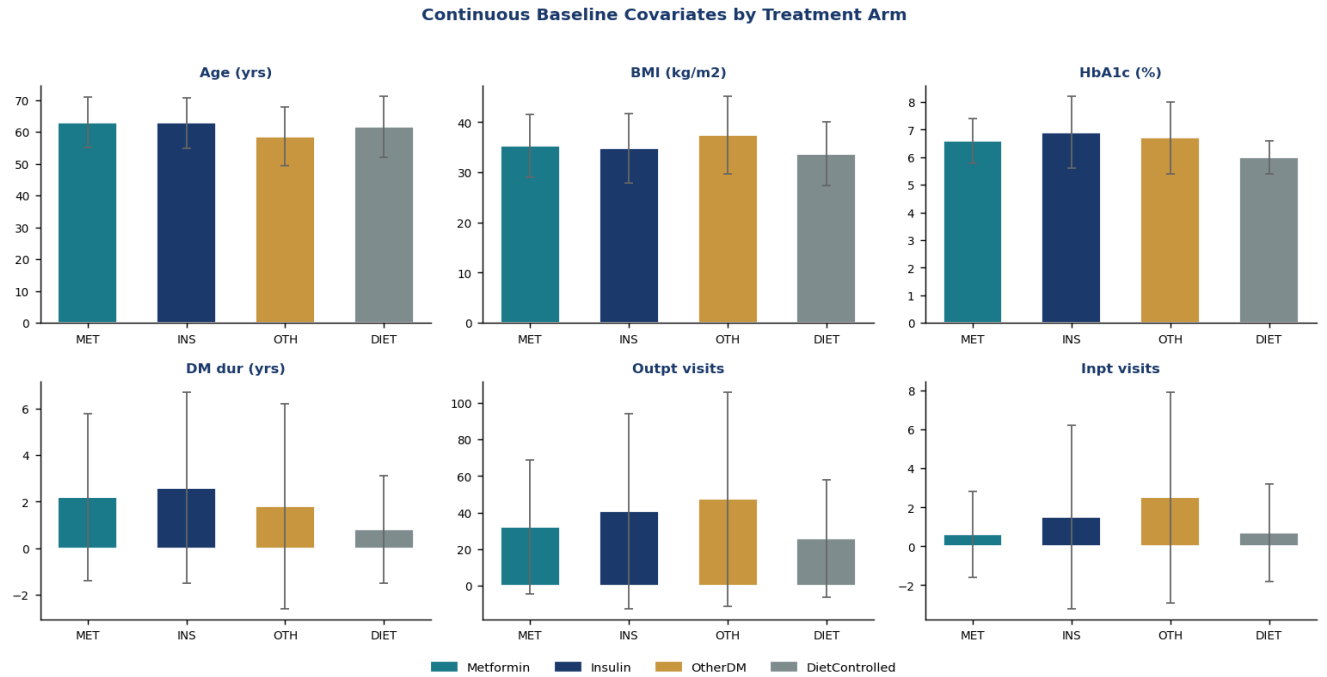


Figure 2. Continuous baseline covariates (mean $\pm$ SD) by treatment arm.

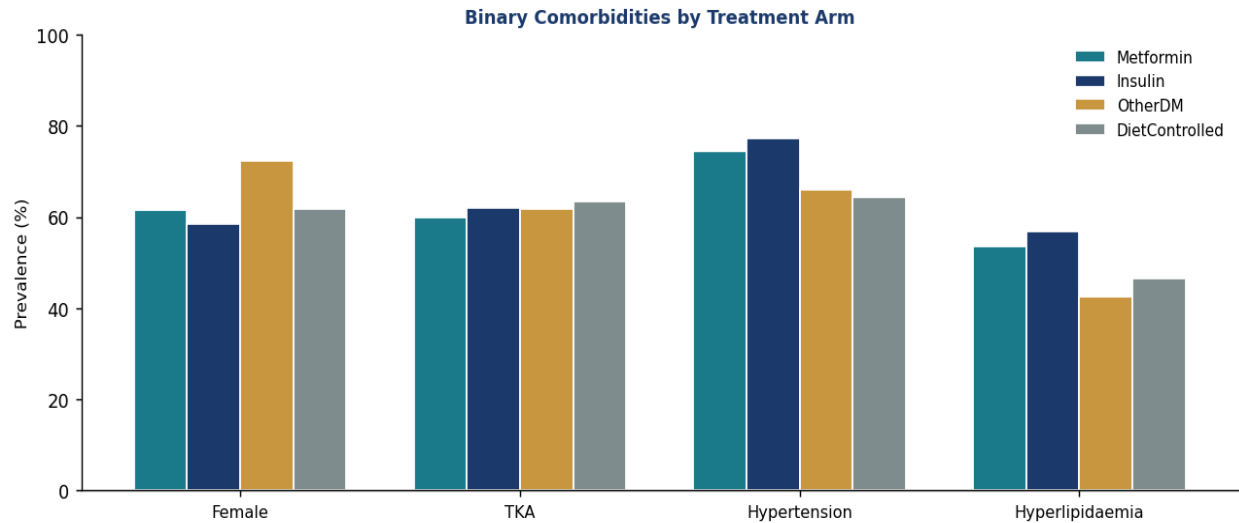


Figure 3. Binary comorbidity and demographic prevalence by arm.

4. Statistical Tests Performed

Every model and sensitivity analysis is described below with its target outcome(s) and the methodological rationale for its selection.

Model / Test	Outcomes Tested	Rationale
Firth Penalised Logistic (primary, bootstrap CI)	readmit_90d readmit_30d readmit_or_ed_90d	Addresses separation bias in small-arm analyses. Bootstrap CIs (n=1,000) preferred over asymptotic when arm sizes are unequal.
GLM with HC3 Robust SE (cross-check)	readmit_90d readmit_30d readmit_or_ed_90d	Independent validation with heteroskedasticity-consistent SEs. Concordance between Firth and GLM increases confidence.
8A: 180-day exposure window	readmit_90d readmit_30d	Tests whether the 365-day arm assignment assumption drives the findings. If estimates are stable, the window choice is not an artefact.
8B: Subgroup by surgery type (THA / TKA)	readmit_90d	THA and TKA have different complication profiles and recovery arcs. Stratification tests whether arm effects are procedure-specific or generalise across both.
8C: Active pharmacotherapy only (excl. DietControlled)	readmit_90d readmit_30d	Removes the heterogeneous DietControlled group to isolate the drug-class contrasts among patients on confirmed pharmacological treatment.

5. Table 1 — Baseline Characteristics

Baseline characteristics by treatment arm. The OtherDM arm has the highest mean BMI (37.4 kg/m²) and the greatest inpatient visit burden (2.5), while DietControlled patients have the shortest diabetes duration (0.8 yrs) and lowest HbA1c (6.0%), consistent with milder or earlier-stage disease.

Variable	Metformin (n=125)	Insulin (n=197)	OtherDM (n=47)	DietCtrl (n=607)
Age (yrs)	63.0 ± 8.0	62.8 ± 7.9	58.5 ± 9.2	61.6 ± 9.5
BMI (kg/m ²)	35.3 ± 6.3	34.8 ± 6.9	37.4 ± 7.7	33.7 ± 6.4
HbA1c (%)	6.6 ± 0.8	6.9 ± 1.3	6.7 ± 1.3	6.0 ± 0.6
DM Duration (yrs)	2.2 ± 3.6	2.6 ± 4.1	1.8 ± 4.4	0.8 ± 2.3
Outpatient visits	32.1 ± 36.8	40.7 ± 53.2	47.3 ± 58.4	25.8 ± 32.1
Inpatient visits	0.6 ± 2.2	1.5 ± 4.7	2.5 ± 5.4	0.7 ± 2.5
ED visits	0.6 ± 2.5	0.8 ± 2.0	0.2 ± 0.5	0.5 ± 1.8
Female	77 (61.6%)	115 (58.4%)	34 (72.3%)	375 (61.8%)
TKA	75 (60.0%)	122 (61.9%)	29 (61.7%)	385 (63.4%)
Hypertension	93 (74.4%)	152 (77.2%)	31 (66.0%)	390 (64.3%)
Hyperlipidaemia	67 (53.6%)	112 (56.9%)	20 (42.6%)	282 (46.5%)
— OUTCOMES —				
90-Day Readmit or ED	24 (19.2%)	65 (33.0%)	21 (44.7%)	172 (28.3%)

6. Unadjusted Event Rates

Raw event counts and rates before covariate adjustment. OtherDM consistently shows the highest rates across all three outcomes.

Outcome	Arm	Events	N	Rate
90-Day Readmission (co-primary)	DietControlled	123	607	20.3%
	Insulin	42	197	21.3%
	Metformin	17	125	13.6%
	OtherDM	19	47	40.4%
30-Day Readmission (co-primary)	DietControlled	95	607	15.7%
	Insulin	28	197	14.2%
	Metformin	11	125	8.8%
	OtherDM	15	47	31.9%
90-Day Readmit or ED (secondary)	DietControlled	172	607	28.3%
	Insulin	65	197	33.0%
	Metformin	24	125	19.2%
	OtherDM	21	47	44.7%

Unadjusted Event Rates by Arm

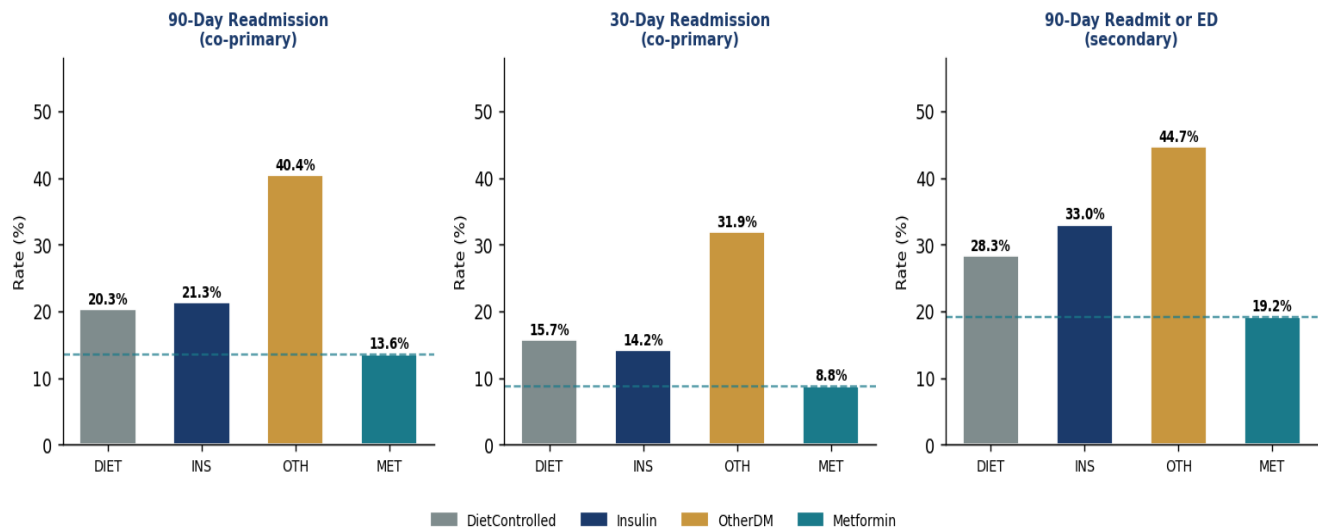


Figure 4. Unadjusted event rates. Dashed line = Metformin reference rate.

7. Adjusted Analysis (Primary & Secondary Outcomes)

Adjusted odds ratios from Firth penalised logistic regression (primary) and GLM HC3 (cross-check). **Bold red** = statistically significant ($p < 0.05$). Reference = Metformin.

Comparison	Outcome	Firth aOR [95% CI]	p	GLM aOR [95% CI]	p	Sig?
Insulin vs MET	90-Day Readmit	1.55 [0.85-3.20]	0.160	1.59 [0.83-3.04]	0.163	No
	30-Day Readmit	1.43 [0.68-3.31]	0.308	1.46 [0.68-3.14]	0.337	No
	90-Day Rdmt+ED	1.91 [1.08-3.79]	0.018	1.96 [1.09-3.50]	0.024	YES
OtherDM vs MET	90-Day Readmit	3.55 [1.63-9.20]	0.002	3.88 [1.64-9.21]	0.002	YES
	30-Day Readmit	3.83 [1.67-11.14]	0.002	4.33 [1.65-11.38]	0.003	YES
	90-Day Rdmt+ED	3.21 [1.50-7.95]	0.004	3.48 [1.49-8.13]	0.004	YES
DietCtrl vs MET	90-Day Readmit	1.53 [0.92-3.09]	0.094	1.54 [0.89-2.70]	0.126	No
	30-Day Readmit	1.76 [1.07-4.00]	0.030	1.78 [0.96-3.30]	0.066	Firth only
	90-Day Rdmt+ED	1.44 [0.90-2.60]	0.130	1.44 [0.87-2.38]	0.155	No

Key findings: OtherDM shows the largest and most consistent associations (aOR 3.21-3.83, $p \leq 0.004$ across all outcomes). Insulin is significant only for the secondary composite (aOR 1.91, $p=0.018$). DietControlled has borderline evidence for 30-day readmission by Firth only (aOR 1.76, $p=0.030$; GLM $p=0.066$). Firth and GLM estimates are concordant in direction and magnitude throughout.

Adjusted Odds Ratios – Firth Penalised Logistic (ref = Metformin)

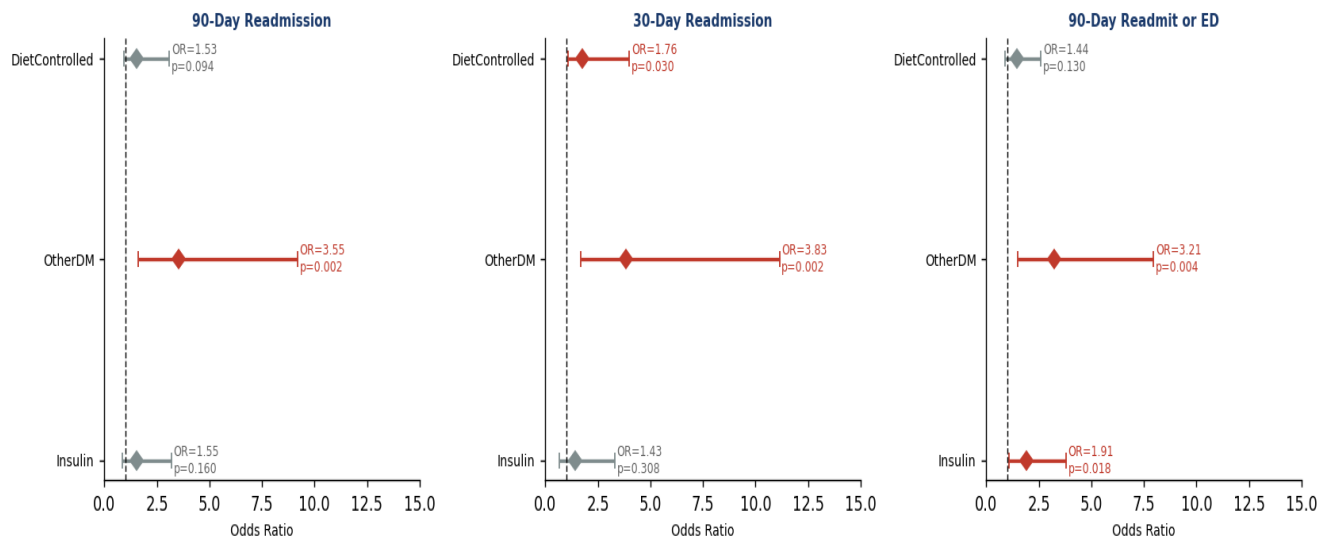


Figure 5. Forest plots of adjusted ORs. Red = significant ($p < 0.05$). Dashed vertical = OR of 1.

8. Sensitivity Analyses

8A — 180-Day Exposure Window

All primary findings replicated with a narrower 180-day look-back. OtherDM remained significant for both co-primary outcomes. Insulin and DietControlled remained non-significant. Point estimates were within the bootstrap CI of the primary analysis.

Comparison	Outcome	aOR [95% CI]	p	Sig?
Insulin vs MET	90-Day Readmit	1.60 [0.84-3.48]	0.132	No
Insulin vs MET	30-Day Readmit	1.44 [0.69-3.41]	0.306	No
OtherDM vs MET	90-Day Readmit	3.41 [1.62-9.49]	0.004	YES
OtherDM vs MET	30-Day Readmit	3.50 [1.43-11.29]	0.016	YES
DietCtrl vs MET	90-Day Readmit	1.49 [0.87-2.92]	0.138	No
DietCtrl vs MET	30-Day Readmit	1.57 [0.93-3.23]	0.088	No

8B — Subgroup by Surgery Type

The OtherDM signal concentrates in TKA (aOR 3.89, p=0.006) while THA does not reach significance (aOR 2.40, p=0.210), likely reflecting reduced power (n=18 OtherDM in THA). No other subgroup comparisons reach significance.

Surgery	Comparison	90-Day Readmit aOR [95% CI]	p	Sig?
THA	Insulin vs MET	1.10 [0.39-2.92]	0.866	No
THA	OtherDM vs MET	2.40 [0.64-10.53]	0.210	No
THA	DietCtrl vs MET	1.23 [0.60-3.46]	0.536	No
TKA	Insulin vs MET	1.79 [0.86-4.67]	0.114	No
TKA	OtherDM vs MET	3.89 [1.52-15.72]	0.006	YES
TKA	DietCtrl vs MET	1.84 [0.96-4.24]	0.062	No

8C — Active Pharmacotherapy Only

Excluding DietControlled patients and comparing only drug-treated arms confirms all primary conclusions. OtherDM remains significant (p=0.002), Insulin non-significant.

Comparison	Outcome	aOR [95% CI]	p	Sig?
Insulin vs MET	90-Day Readmit	1.55 [0.85-3.20]	0.160	No
Insulin vs MET	30-Day Readmit	1.43 [0.68-3.31]	0.308	No
OtherDM vs MET	90-Day Readmit	3.55 [1.63-9.20]	0.002	YES
OtherDM vs MET	30-Day Readmit	3.83 [1.67-11.14]	0.002	YES

9. Results Summary

Outcome	MET rate	INS rate	OTH rate	DIET rate	aOR Insulin [CI] p	aOR OtherDM [CI] p	aOR Diet [CI] p
90-Day Readmit (co-primary)	17/125 (14%)	42/197 (21%)	19/47 (40%)	123/607 (20%)	1.55 [0.85-3.20] p=0.160 ✗	3.55 [1.63-9.20] p=0.002 ✓	1.53 [0.92-3.09] p=0.094 ✗
30-Day Readmit (co-primary)	11/125 (9%)	28/197 (14%)	15/47 (32%)	95/607 (16%)	1.43 [0.68-3.31] p=0.308 ✗	3.83 [1.67-11.14] p=0.002 ✓	1.76 [1.07-4.00] p=0.030 ~
90-Day Rdmt+ED (secondary)	24/125 (19%)	65/197 (33%)	21/47 (45%)	172/607 (28%)	1.91 [1.08-3.79] p=0.018 ✓	3.21 [1.50-7.95] p=0.004 ✓	1.44 [0.90-2.60] p=0.130 ✗

✓ = $p < 0.05$ both methods ~ = Firth significant, GLM borderline ✗ = not significant

10. Outcome Definitions — Clarification

Question: Are 90-day readmissions (co-primary) admissions directly related to the surgery, vs the 90-day readmission or ED (secondary)?

In this EHR/claims-based study, **90-day readmission is all-cause** — it captures *any* inpatient hospital admission occurring within 90 days of the index THA/TKA, whether directly related to the surgical procedure (e.g., wound infection, prosthetic dislocation, pulmonary embolism, fracture) or not (e.g., uncontrolled diabetes, cardiac event, urinary tract infection). This is the standard approach in administrative/EHR-based arthroplasty outcomes research because reliably attributing admissions to a specific surgical cause requires additional adjudication beyond what routine claims data permit.

The **90-day readmission or ED visit (secondary composite)** widens the capture net by adding all emergency department visits that did *not* result in an inpatient admission. This is clinically important because patients with post-surgical complications or uncontrolled DM may present to the ED, receive treatment, and be discharged without crossing the threshold for admission — events that would be completely invisible to the readmission-only endpoint. The composite therefore captures the *full unplanned acute-care utilisation burden* following surgery, while the co-primary readmission outcomes measure only the most severe tier of that utilisation requiring hospital-level care.

Outcome	What it captures	Severity tier	Relation to surgery
90-Day Readmission (co-primary)	Any inpatient admission $\leq$ 90d post-surgery (all-cause)	High — requires hospital admission	Includes surgery-related AND unrelated causes. Cannot be restricted to surgical causes without manual adjudication in EHR data.
30-Day Readmission (co-primary)	Any inpatient admission $\leq$ 30d post-surgery (all-cause)	High — requires hospital admission	Narrower window; higher proportion likely surgery-related but still all-cause by definition.
90-Day Readmit or ED (secondary)	Any inpatient admission OR any ED visit $\leq$ 90d post-surgery	Moderate to High — captures near-miss events	Broader composite. Adds ED visits that did not result in admission. Represents full unplanned acute-care burden.

11. Study Limitations

This study has several limitations that should be considered when interpreting findings and designing future research.

Limitation	Description	Impact on Interpretation
Retrospective observational design	Arm assignment is based on pre-existing drug dispensation, not randomisation. Residual confounding from unmeasured variables (surgeon volume, implant type, anaesthesia, pre-operative physical therapy) cannot be excluded.	Cannot establish causality. Associations should be interpreted as hypothesis-generating rather than definitive.
Small OtherDM arm (n=47)	The OtherDM arm has limited power, particularly in subgroup analyses (e.g., n=18 in THA). Wide confidence intervals reflect this uncertainty.	Significant OtherDM findings should be interpreted with caution. Replication in larger cohorts is essential.
Incomplete HbA1c availability (47%)	HbA1c was available for only 458/976 patients, limiting the ability to fully adjust for glycaemic control at the time of surgery.	Residual confounding by pre-operative glycaemic status is possible, particularly for the OtherDM and Insulin arms.
Incomplete BMI availability (55%)	BMI was missing for 444/976 patients. Available-case analysis was used.	Potential bias if missingness is related to arm assignment or outcomes.
All-cause readmission endpoint	The readmission outcome captures any hospital admission within the window, not just surgery-related admissions. This may dilute the true surgical complication signal.	Associations may partly reflect differences in general health status rather than surgery-specific complications.
GLP-1 agonists merged into OtherDM	GLP-1 receptor agonists (n=218) were not isolated as a separate arm due to their distribution across other arms. Potential differential effects of GLP-1 agents on readmission cannot be assessed.	The OtherDM effect estimate is a composite of multiple drug classes with potentially different mechanisms.
Single-database study	Results are derived from one EHR database and may not generalise to other healthcare systems, payer types, or geographic regions.	External validity should be assessed before clinical implementation of these findings.
No surgical complication granularity	The study cannot distinguish between readmissions for prosthetic complications (dislocation, infection, fracture) and those for medical causes (DM decompensation, cardiac events).	Future studies should use procedure-specific readmission codes to isolate the surgical complication pathway.

12. Journal Submission: Journal of Arthroplasty

The Journal of Arthroplasty (Elsevier) is the leading peer-reviewed journal focused on surgical and clinical outcomes of total hip and knee replacement. It is the primary target for this readmission study given its scope, audience (orthopaedic surgeons and researchers), and impact factor. The table below summarises the submission requirements, costs, and strategic considerations.

Category	Details	Action Required
Journal	Journal of Arthroplasty Publisher: Elsevier ISSN: 0883-5403 (print), 1532-8406 (online)	Confirm current author guidelines at www.arthroplastyjournal.org
Impact Factor	Approximately 3.8-4.0 (2024 JCR) CiteScore: ~6.1 Q1 in Orthopaedics	Competitive submission; well within scope for outcomes research.
Article Type	Original Research Article (clinical/outcomes) Alternative: Research Letter if word count restricted	Select 'Original Research' in submission portal.
Word Limit	3,500 words maximum (excluding abstract, references, tables, figures) Abstract: 250 words, structured (Background/Methods/Results/Conclusions)	Draft manuscript to fit 3,500-word body. Structured abstract required.
Abstract Format	Structured: Background, Methods, Results, Conclusions Maximum 250 words No citations in abstract	Ensure all four sections present. Include key aOR values and p-values in Results section.
Tables & Figures	Up to 6 figures and 6 tables total All figures in TIFF/EPS, minimum 300 DPI Tables in editable format (Word)	Include: Table 1 (baseline), results table, sensitivity summary. Figures: forest plots, outcome rate charts.
References	30-40 references recommended Vancouver (numbered) format All citations must be accessible	Cite readmission benchmarks, Firth regression methodology, and prior DM-arthroplasty literature.
Checklist Required	STROBE checklist (observational studies) Conflict of interest declarations for all authors IRB / Ethics approval statement Data availability statement	Complete STROBE checklist before submission. Attach as supplementary file. Confirm IRB approval language.
Open Access Options	Option A: Subscription (no APC) — available to all subscribers Option B: Open Access (Gold) — Article Processing Charge (APC): approximately USD 3,900 Option C: Author Accepted Manuscript posted to PubMed Central after 12-month embargo	Confirm funding availability if open access desired. NIH-funded studies may require immediate open access.
Submission Fee	No submission fee	No upfront cost to submit.
Peer Review Timeline	Initial decision: approximately 4-6 weeks Revision turnaround: 4-8 weeks Final decision after revision: 2-4 weeks Total time to acceptance: 3-6 months	Prepare a detailed response-to-reviewers document. Expect requests to expand sensitivity analyses or add clinical context.
Key Submission Tips	1. Emphasise clinical relevance for orthopaedic surgeons 2. Frame OtherDM finding as a pre-operative risk stratification tool 3. Address GLP-1 limitation prominently — this is a high-interest area 4. Include absolute risk differences alongside ORs for clinical interpretability	Consider adding a 'Clinical Relevance' box — a journal requirement for some article types. Check current author instructions.
Competing Journals	Alternatives if rejected: - Arthroplasty Today (Elsevier, open access, APC ~USD 1,500) - Journal of Bone & Joint Surgery (JBJS) - Bone & Joint Journal - Clinical Orthopaedics and Related Research (CORR)	Prepare a tiered submission strategy before initial submission.

Method: Firth penalised logistic regression with bootstrap 95% CI (1,000 iterations). Cross-checked with GLM HC3 robust SE. Covariates: age, BMI, HbA1c, diabetes duration, surgery type, HTN, HLD, CAD, depression, baseline inpatient visits. GLP-1 receptor agonists merged into OtherDM. Reference: Metformin.